T.C.
YILDIZ TEKNİK ÜNİVERSİTESİ
İKTİSADİ VE İDARİ BİLİMLER FAKÜLTESİ
İKTİSAT PROGRAMI

COMPARISON OF PROBABILITY OF DEFAULT MODELS USED IN CREDIT RISK MANAGEMENT

BURAK UZUN
18035028

TEZ DANIŞMANI
Prof. Dr. HÜSEYİN TAŞTAN

İSTANBUL
2022

T.C
YILDIZ TECHNICAL UNIVERSITY
THE FACULTY OF ECONOMICS AND ADMINISTRATIVE SCIENCES
DEPARTMENT OF ECONOMICS

BACHELOR’S THESIS

COMPARISON OF PROBABILITY OF DEFAULT MODELS USED IN CREDIT RISK MANAGEMENT

BURAK UZUN
18035028

THESIS ADVISOR
Prof. Dr. HÜSEYİN TAŞTAN

ISTANBUL
2022

T.C.
YILDIZ TEKNİK ÜNİVERSİTESİ
İKTİSADİ VE İDARİ BİLİMLER FAKÜLTESİ
İKTİSAT PROGRAMI

LİSANS TEZİ

COMPARISON OF PROBABILITY OF DEFAULT MODELS USED IN CREDIT RISK MANAGEMENT

BURAK UZUN
18035028

Tezin Savunulduğu Tarih:
Tez Oy Birliği / Oy Çokluğu ile Başarılı Bulunmuştur

Unvan Ad Soyad:	**İmza**
Tez Danışmanı:	
Jüri Üyeleri:	

İSTANBUL
HAZİRAN 2022

ÖZ

COMPARISON OF PROBABILITY OF DEFAULT MODELS USED IN CREDIT RISK MANAGEMENT

Burak Uzun

Haziran, 2022

Bu çalışmada, beklenen kredi zararı ve kredi riskinin hesaplanmasında kullanılan temerrüt oranı modelleri, metodlar bağlamında karşılaştırılmıştır. Bu karşılaştırma, oldukça maliyetli olan ve Weight of Evidence gruplarıyla çalışan lineer modeller ve sınıflar arası bölünmeyi otomatik olarak gerçekleştiren ağaç tabanlı makine öğrenmesi yöntemleri arasında yapılmıştır. Ağaç tabanlı makine öğrenimi modellerinde XGBoost, Light GBM, Cat Boost, Random Forest ve Klasifikasyon ve Regresyon Ağaçları (CART) kullanılmıştır. Kullanılan veri seti bir taraflararası borç verme şirketi olan Lending Club'un 2007-2014 yıllarına ait kredi verisidir. Bu veri seti bulundurduğu temerrüt oranı ve gözlem sayısı açısından model performansı kıyaslamaya uygun bulunduğu için tercih edilmiştir. Veri setinden Bilgi Değeri (Information Value) değerlerine göre seçilen 22 değişken hem lineer modeller de hem de ağaç tabanlı makine öğrenimi modellerinde kullanılmıştır. Çalışmamızda AUC, Brier score, Kolmogorov Smirnov, F1 Score, Precision, Recall metrikleri kullanılmıştır. Başarı metrikleri eğitim verisiyle öğrenen modelin, test verisi ile valide edilmesiyle elde edilmiştir. Modellerin ürettiği olasılık tahmin değerlerinde 0.80 üzeri değerler non-default (1) olarak atanmıştır. Hiper-parametre optimizayonunda GridSearchCV ve Optuna algoritmaları kullanılmış ve temel öğrenme hiper-parametreleri optimize edilmiştir. Elde ettiğimiz bulgulara göre, WoE gruplamaları temerrüte düşen kredileri tahmin etmek adına daha başarılı bulunmuştur. Öte yandan kullandığımız metodlar arasından XGBoost, WoE gruplamaları ile en iyi temerrüt tahmin etme sonucunu vermiştir. Elde ettiğimiz bulgulara göre XGBoost modeli geleneksel modellere rakip olabilecek kapasitede bulunmuştur. Bankaların kredi riski hesaplamasında uygulanan regülasyonlar makine öğrenmesi metodlarının açıklanabilirlik ve güvenilirlik açısından hala yetersiz bulmakta olduğu için bu çalışma geleneksel metodlardan makine öğrenimi metodlarına geçiş sürecinde tahmin gücü ve maliyet verimliliği kazanımlarına dikkat çekiyor.

Anahtar Kelimeler: Kredi Riski, Risk Yönetimi, Makine Öğrenmesi, PD, EAD, LGD, Model Karşılaştırması

ABSTRACT

COMPARISON OF PROBABILITY OF DEFAULT MODELS USED IN CREDIT RISK MANAGEMENT

Burak Uzun

June, 2022

In this study, various probability of default models used in the calculation of credit risk are compared in the context of methods. Linear models that work with Weight of Evidence groups, which are quite costly, and tree-based machine learning methods that automatically perform the division between classes are compared. XGBoost, Light GBM, Cat Boost, Random Forest and Classification and Regression Trees (CART) are used in machine learning models. The data set used is the loan data of Lending Club, a peer-to-peer lending company, covering the years 2007-2014. This data set was preferred because it is satisfactory in terms of default rate and number of observations. 22 variables that are selected from the data set according to their Information Value values are used in both linear models and tree-based machine learning models. In our study, metrics such as AUC, Brier score, Kolmogorov Smirnov were used. Success metrics were obtained by validating the model with the test data. In the probability estimation values produced by the models, values above 0.80 are assigned as non-default (1). GridSearchCV and Optuna algorithms were used for hyper-parameter optimization and basic hyperparameters were optimized. According to our findings, WoE groupings are very successful in predicting defaulters. On the other hand, among the methods we used, XGBoost gave the best results with WoE groupings. We can say that XGBoost is capable of challenging traditional models. The harsh regulations applied in the calculation of credit risk of banks still find machine learning methods insufficient in terms of explainability and reliability. This study highlights the predictive power and cost efficiency gains in the transition from traditional methods to machine learning methods.

Key Words: Credit Risk, Risk Management, Machine Learning, PD, EAD, LGD, Model Comparison

ACKNOWLEDGEMENTS

I would like to thank my thesis advisor, Prof. Dr. Hüseyin Taştan who supported me during the preparation of this study and introduced me to the field of machine learning.

Istanbul, Haziran, 2022 Burak Uzun

INDEX

LIST OF TABLES

LIST OF FIGURES

ABBREVIATIONS

AIC	: Akaike Information Criterion
AUC	: Area Under Curve
BIC	: Bayesian Information Criterion
BIS	: Bank for International Settlements
CATBOOST	: Categorical Boosting
EAD	: Exposure at Default
ECL	: Expected Credit Loss
FDIC	: Federal Deposit Insurance Corporation
GPU	: Graphics Processing Unit
IV	: Information Value
KS	: Kolmogorov Smirnov
LASSO	: Least Absolute Shrinkage and Selection Operator
LGD	: Loss Given Default
P2P	: Peer-to-Peer Lending
PD	: Probability of Default
ROC Curve	: Receiver Operating Characteristics Curve
RWA	: Risk Weighted Assets
SMBO	: Sequential Model-Based Optimization
The Fed	: Federal Reserve Board
WoE	: Weight of Evidence
XGBoost	: Extreme Gradient Boosting

1. INTRODUCTION

Risk management, in its broadest terms, is the act of identifying, assessing, and controlling any threaths that might adversely affect a company's growth or sustenance. In other words, it is the discipline that is concerned with the logical framework, and the implementations of an action plan resulting from this framework, when dealing with a potential risk. This is a term applied mostly to finance and banking as the most common practices of risk management present themselves with the aim of avoiding the loss of capital and increasing the value of assets. Georges Dione, in his paper "Risk Management: History, Definition and Critique", defines risk management as "a set of financial or operational activities that maximize the value of a company or a portfolio by reducing the costs associated with cash flow volatility". [1]

Risk management offfers two critical advantages for banks. The first being the sustenance aspect by protecting the bank from possible losses, and the second being the growth aspect by determining the risks beforehand and taking mandatory precautions. Poor risk management can have severe effects nationally and globally as examplified by the 2008 subprime crisis.

Risk management in banks primarily begins with identifying risks measuring them. Banks build their risk management framework on numerical expressions that are decided based on certain criteria of the risks that the bank is exposed to. Then, they use these measurements to calculate risk wieghted assets.

1.1. Risk Weighted Assets and Capital Requirements

The major risks that banks face are credit risk, market risk, operational risk, and liquidity risk. Credit risk is the possibility of borrowers or counterparties failing to meet their repayment obligations (defaulting), market risk deals with the threats arising from the volatility of the market, operational risk covers the failures of a bank's internal processes, and liquidity risk refers to a bank's abilities of meeting funding obligations. Since lending loans is one of the basic functions of classical banking,

[1] Dionne, G. (2013). *Risk Management: History, Definition, and Critique.* ERN: Other Microeconomics: Decision-Making under Risk & Uncertainty.

credit risk is the first defined risk in the banking literature. We can argue that the biggest risk banks face is also credit risk as the inability of a borrower to meet their obligations can cause severe interruptions of cash flow.

Banks always face the risk of borrowers defaulting or investments flatlining, no matter how strong the precautions are, they preserve a certain amount of their capital in order to prevent insolvency. This minimum amount is referred to as risk weighted assets (RWA). Similarly capital requirements are regulations for banks which determine how much capital a bank has to hold. These requirements are regulated by agencies such as the Bank for International Settlements (BIS), the Federal Deposit Insurance Corporation (FDIC), or the Federal Reserve Board (the Fed).

1.2. Calculating Credit Risk

For banks, lending allows room for high profits but this possibility comes at the expense of risk taking. So they must evaluate the risk-return relationship for each loan, and they must calculate how much capital they have keep in the reserve for a taken risk as discussed in section 1.1 of this paper. In order to determine the risk weighted assets banks have to calculate expected credit loss (ECL.)

"Expected Credit Loss is the probability-weighted estimate of credit losses (i.ei the present value of all cash shortfalls) over the expected life of a financial instrument. A cash shortfall is the difference between the cash flows that are due to an entity in accordance with a contract (the scheduled or contractual cashflows) and the cash flows that the entity expects to receive (the actual expected cashflows). Given that expected credit losses consider both the amount and timing of payments, a credit loss arises even if the entity expects to be paid in full but later than when contractually due."[2]

The expected loss is calculated in quantitative terms as follows:

$$\mathbf{ECL = PD \times LGD \times EAD}$$

Formula 1

where

[2] Expected Credit Loss. (2022, February 18). Open Risk Manual, . Retrieved 23:41, June 5, 2022 from https://www.openriskmanual.org/wiki/index.php?title=Expected_Credit_Loss&oldid=27765.

PD = probability of default

LGD = loss given default

EAD = exposure at default

The aforementioned terms will be discussed in detail in the following section.

Legal regulations still require logistic regression in probability of default calculations for institutions that want to calculate their own internal credit risk. Because of that, in this study, we will focus on the comparison of modeling methods (conventional/legal models and tree based machine learning models) used for calculating probability of default. In order to do that, proper independent variables will be selected by using WoE groupings and Information Values. These independent variables will be used in both linear and tree based models; but because WoE groupings and logistic regression are the currently used methods in the banking sector, our comparison's main aim is using the data that is not divided into WoE groups on tree based ML models. Of course, the WoE grouping method will also be used for the variables that include the missing values on ML data. Therefore, individual PD models will be created in this study to make this comparison.

We will be making observations regarding the modeling methods used for calculating probability of default because, as it was mentioned in the above paragraphs, it is a legal obligation for financial institutions to calculate their expected credit loss Then, if better models can be developed, the lending process will be be more efficient.

2. EXPECTED CREDIT LOSS COMPONENTS

In this section, PD, LGD and EAD models will be briefly explained.

2.1. Probability of Default

Probability of Default is the probability that a person or institution that borrows from a bank will not repay or be unable to pay its debt. This probability calculation is made by building models with customer data to estimate the likelihood of a being default. Because big financial institutions deal with millions of customers, it's not easy to build a succesful predictive model and keep it sustainable.

2.2. Exposure At Default

Exposure At Default (EAD) is an estimation of the amount of loss that the bank will face in the case of the customer not repaying or not being able to repay the loan. exposure at default should be updated every time that the borrower makes a payment to the lender/bank. Because exposure at default has a big impact on calculation of risk weighted assets, banks should also calculate their exposure carefully.

2.3. Loss Given Default (LGD)

Loss Given Default (LGD) is the value that represents how much of a defaulted loan would be lost. Loss given default simply corresponds (1 – Recovery Rate). To calculate loss given default in terms of dollars, we can use the following simple equation, which is,

$$LGD(\$) = Exposure\ At\ Default * (1 - Recovery\ Rate)$$

Formula 2

3. DATA

3.1. Peer-to-peer Lending

P2P (Peer-to-peer) lending, unlike traditional bank loans, connects borrowers directly to lenders, without a financial institution -such as a bank- acting as the intermediary. This online intermediary, which first emerged in 2005 in the United Kingdom with the platform Zopa, is favorable for individuals and small businesses as it has been more difficult for these groups to obtain traditional bank loans since the financial crisis in 2008. (Tang et al, 2020) Banks have chosen to credit ration after the high default rates, which resulted from subprime lending, brought about a mortgage crisis.

Because P2P platforms operate through online networks, the intermediation costs are lower compared with their traditional counterpart due to the absence of branches. As it is mentioned, the first P2P platform was Zopa in 2005. Data shows that since its establishment, P2P lending has been growing significantly. Between 2010 and 2015 the compound annual growth rate of P2P lending in the UK and USA was 151%; additionally, this rate in continental Europe from 2012 to 2014 was 113%. (Polena, 2017). This growth can be linked to the advantageous conditions P2P lending offers. The average interest rates are lower for borrowers, and lenders that hold a well-diversified loan portfolio are awarded higher earnings.

3.2. The Structure and Preprocessing of the Dataset

The dataset used in this paper is called “Lending Club Loan Dataset” and it includes the loan records of the Lending Club, which is a peer-to-peer lending company based in United States. The dataset includes 466285 observations and 75 variables of the individual loan applications between the years 2007 and 2014.

In this dataset; the assignment of dependent variable, which is “loan_status”, will be either default or non-default. Unlike other studies, 1 represents good credit (non-default) while the 0 represents bad credit (non-default). A payment delay of more than 30 days is also accepted as bad credit (default), therefore there are 415317 good and 50968 bad credits defined in the dataset, and that is equivalent to a 10.9% default rate in total. If we were to compare this with real life scenarios this default rate is quite high; but compared to other popular credit datasets (German Credit Dataset, Australian Credit Dataset, Taiwan Credit Dataset are all have less than 1000 observations, and their default rate is higher than 30% in said datasets), Lending Club Loan data will prove to be very useful for this study.

The dataset is splitted as “train set” from January 2007 to 2014 June and as

Figure 1: Distribution of Loan Status

“out-of-sample validation set” from June 2014 to December 2014, which corresponds to 70% of the complete dataset being the train set and the other %30 being the out-of-sample validation set.

3.3. Weight of Evidence Groupings

“Weight of Evidence” or WoE groupings are applied to show a given variable’s power of distinguishing between 2 or more classes according to the values of said variable, in terms of credit risk these classes are default and non-default. This binning process is done by applying fine classification (equal binning) and coarse classification (dispersed binning) to the variables. When the categorical variables are directly passed through the "Coarse Classing" process according to their WoE values, continuous variables are subjected to coarse classing after the binning with fine-classing method. In credit risk modelling this is a time consuming procedure most of the time, but it's also crucial. The need for a credit risk model that can maintain it's predictive power in the long term makes the process of binning necessary.

Despite it's predictive power, WoE grouping can also provide consistent penalizing effects to a bank's credit portfolio as the restrictions decrease the number of unique classes. For instance, if we take the variable that holds the customers' age into consideration, there exists 60 classes between the ages of 20-79 without WoE groupings. However, if we decide on five year intervals to divide this data, there would be 12 unique classes. WoE groupings also convert missing data into another categorical variable which is quite useful when dealing with missing data. We can calculate WoE value in the formula shown below:

$$\boldsymbol{WoE = \left[ln\left(\frac{Frequency\ of\ Non - Defaults}{Frequency\ of\ Defaults}\right)\right] * 100}$$

Formula 3

According to the equation above, we can say that if the Weight of Evidence value is negative for a certain group, the bad observations exceed the good ones in this group and this can be applied vice versa.

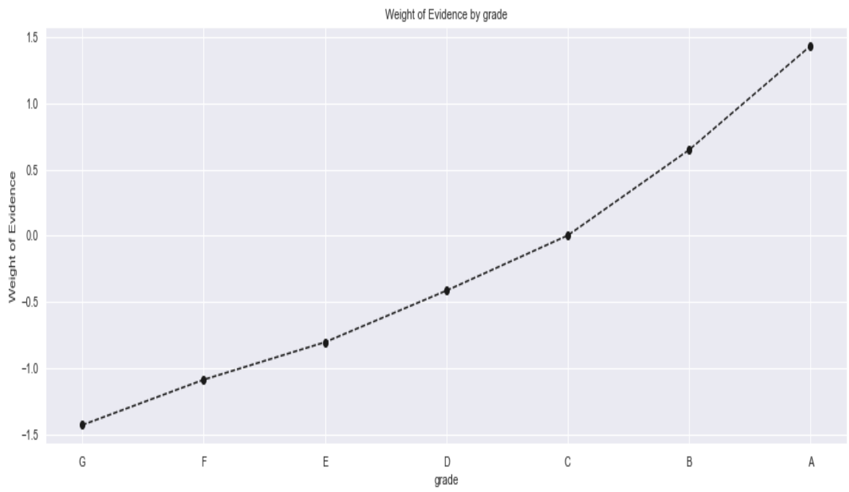


Figure 2: WoE Values for Variable "grade"

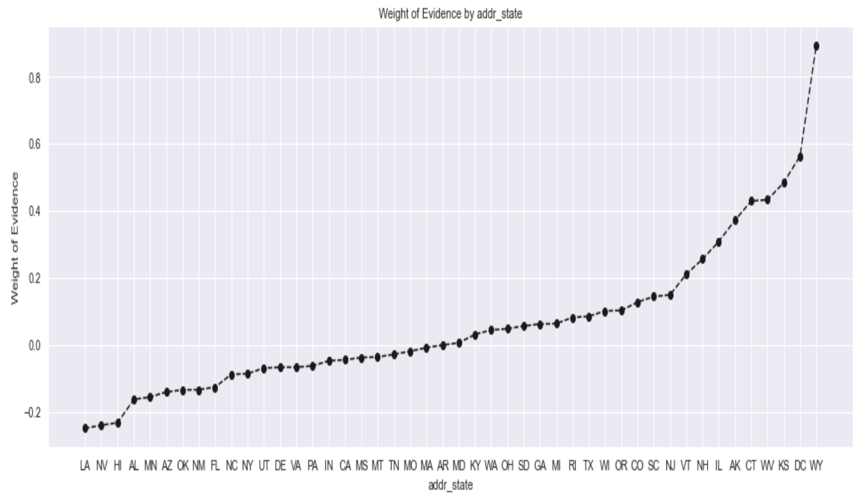


Figure 3: WoE Values for Variable “addr_state”

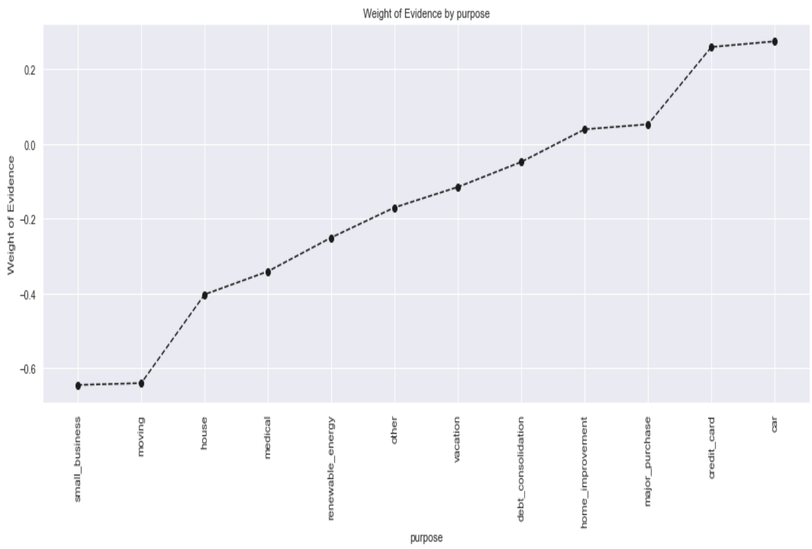


Figure 4: WoE Values for Variable "purpose"

3.4. Information Value (IV)

Just like the Weight of Evidence calculation, Information Value (IV) measures the variable's ability to distinguish between binary dependent variable classes (0,1). The Information Value calculation is applied to each Weight of Evidence class to observe their predictive power. The cumulative summation of these groups IV is called “Cumulative Information Value” and this is the value that represents the distinguishing power of the binned variable. We can calculate the Information Value in the formula shown below:

$$IV = \sum (\boldsymbol{Freq.of\ Non\ Default} - \boldsymbol{Freq.of\ Default}) * \boldsymbol{WoE}$$

Formula 4

The Information Value is very practical because it also helps the "Feature Selection" process in classification problems and especially in Logistic Regression which is the conventional, and also current, method for creating Probability of Default models in the banking sector due to the legal regulations of governmental institutions. To decide which variable should be used for modeling, the main guide for the critical values for IV is as follows:

Information Value	**Predictive Strength**
<= .02	**Insignificant Predictive Power**
.02<IV<=.10	**Low Predictive Power**
.10<IV<=.30	**Medium Predictive Strength**
.30<IV<=.50	**Strong Predictive Power**
>.50	**Suspicious**

Table 1: Relationship Between Information Value and Predictive Strength

Initial feature selection process has been made according to the values above for Logistic Regression and LASSO Logistic Regression models in this study.

4.METHODS

In this section, the methods used in this study or that need to be understood for a more comprehensive evaluation will be explained. These are Feature Selection methods, Logistic Regression, LASSO, and some supervised machine learning methods.

4.1. Feature Selection

Feature selection is the most optimal selection of the independent variables to be used in estimateing the dependent variable. The main reason why the feature selection process is important is that unsuccessful variable selection can harm the performance of the model, especially in linear models. In models that are not immune to multicollinearity, different variables carrying the same information can lead to inaccurate estimates. In addition, in scenarios where working with very large data is necessary, variables that include the same information or don’t illustrate a relationship with dependent variable (non-informative) may cause not only reduction of model performance, but also a technical burden and therefore it increases the model’s cost in terms of time. (Hild, 2021)

4.1.1. Forward Stepwise Feature Selection

Forward Stepwise Selection is the one of the most widespread approaches in feature selection process. The method gets its name from the fact that it initially takes a single independent variable and adds a new independent variable at each step. This approach, which is not the same but a more comprehensive version of the Forward Selection approach, reexamines all the variables in each new selection. This is a very useful method because when new independent variables are selected in the regression, the previous independent variables may have lost their explanatory power. In this case, the variables with decreasing explanatory power on a certain level are removed from the model.

The Feature Selection process continues until a certain stopping rule is reached. A stopping rule is formed when the variables satisfies a certain threshold level in the variable selection. For a more detailed explanation about these thresholds, AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) can be examined.

4.1.2. Backward Stepwise Selection/Backward Feature Elimination

In the Backward Feature Elimination method, the model is initially set up with all the variables, as opposed to the Forward Stepwise approach. The statistical significance values of the variables are calculated in each step and the variable with the lowest significance level is eliminated from the model. This process continues iteratively until a certain stopping rule is reached.

4.2. Models

4.2.1. Logistic Regression

Logistic regression is the most widely used conventional method in classification problems. It very similar with linear regression, but unlike linear regression, logistic regression calculates the probability of an event occurring or not in order to explain a binary dependent variable. To comprehend this predictive modeling method, the concept of linear regression should be understood . Linear regression with one explanatory variable equation is shown below:

$$\boldsymbol{Y_i = \beta_0 + \beta_1 x_1 + \epsilon_i}$$

Formula 5

We can interpret the equation above Y as a dependent variable, β_0 as a intercept coefficient (constant), β_1 as a coefficent of independent variable x_1 and ϵ_i as error term(unobserved part). To understand how to obtain and optimize these coefficients with the given data, one should be informed about Ordinary Least Squares method. The basic explanation for Ordinary Least Squares is that it is an optimization method to obtain OLS estimator coefficient which can be represented as $\hat{\beta}$.

If we think in a matrix form, equation to compute ordinary least squares estimator would be as follows:

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

Formula 6

This approach minimizes the square of the sum of residuals instead of minimizing the residuals by themselves. The reason is that this prevents the negative and positive residues from canceling each other out in the process of summing up. In the equation above, X is a $n \times k$ vector whic stands for the observations of independent variable and y is a $n \times 1$ vector which stands for the dependent variable. To derive sum of squared residuals we can simply use the given equation:

$$\boldsymbol{SSR} = \left[\sum_{i=1}^{n}(\boldsymbol{y} - \widehat{\boldsymbol{y}})^2\right] = \sum_{i=1}^{n}\widehat{\boldsymbol{u}}^T\widehat{\boldsymbol{u}}$$

Formula 7

Logistic regression explains the log-odds of the binary dependent variable in linear form of the independent variables just like linear regression, but without error term. We can show this equation as follows:

$$\log\left(\frac{\boldsymbol{P}}{1-\boldsymbol{P}}\right) = \boldsymbol{\beta}_0 + \boldsymbol{\beta}_1\boldsymbol{x}_1 + \boldsymbol{\beta}_2\boldsymbol{x}_2 + \ldots + \boldsymbol{\beta}_n\boldsymbol{x}_n$$

If we convert this equation into an equation that takes log(odds) as input variable and gives probability outputs between 0 and 1, we obtain:

$$\boldsymbol{P} = \frac{e^{\boldsymbol{\beta}_0 + \boldsymbol{\beta}_1\boldsymbol{x}_1 + \boldsymbol{\beta}_2\boldsymbol{x}_2 + \cdots + \boldsymbol{\beta}_n\boldsymbol{x}_n}}{1 + e^{\boldsymbol{\beta}_0 + \boldsymbol{\beta}_1\boldsymbol{x}_1 + \boldsymbol{\beta}_2\boldsymbol{x}_2 + \cdots + \boldsymbol{\beta}_n\boldsymbol{x}_n}}$$

Formula 8

To optimize and find the values of this coefficents (β) and fit the model, maximum likelihood estimation approach should be applied. The graph of logistic regression/sigmoid function is shown below.

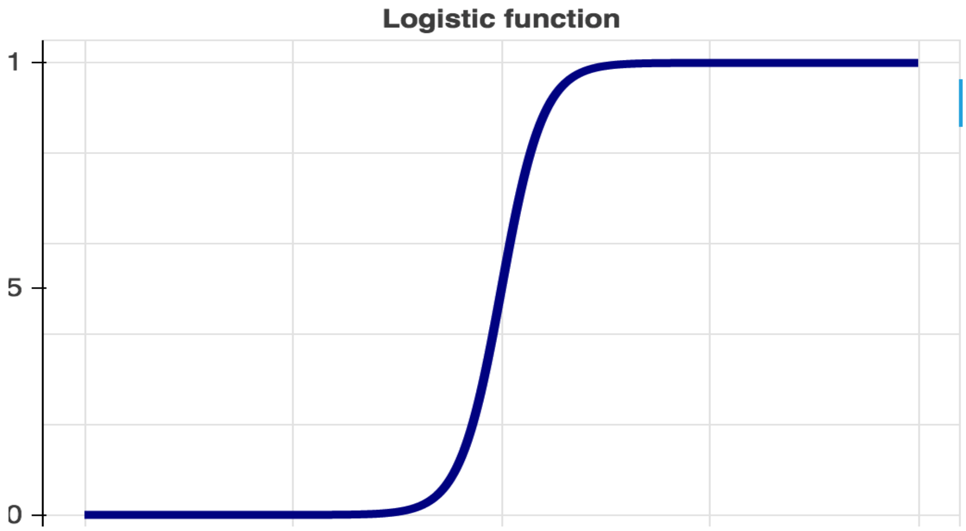


Figure 5: Logistic Function Graph

4.2.2. LASSO Logistic Regression

Least Absolute Shrinkage and Selection Operator (LASSO) is a regulatory/shrinkage method that adds a penalizer term which is equal to the absolute value of the coefficient. This shrinkage technique can cause the elimination of some independent variables/features by setting their coefficients to zero with lasso penalty term. Therefore lasso regularization is very useful for feature selection processes. With the optimization on lambda value, variables with low coefficient values/unimportant features go to zero while variables with high coefficient values shrink to a certain extent. This way, lasso regularization prevents the over-fitting problem and it's also a good way to handle multicollinearity problem. Negative Log – likelihood minimization equation with LASSO penalty term for logistic regression shown below. If the $\beta = (\beta_1, \cdots, \beta_n)$ and $x_i = (x_1, x_2, \ldots, x_n)$ then,

$$\sum_{i=1}^{n} \left[\log\left(1 + e^{\beta_0 + x_i^T \beta}\right) - y_i(\beta_0 + x_i^T \beta) \right] + \lambda \sum_{k=1}^{n} |\beta_k|$$

Formula 9

The term lambda (λ) optimization usually made by using GridSearchCV, which is the method that is also used in this study and we'll also explain this topic in the "Hyperparameter Optimization/Model Tuning" section. When the term lambda (λ)

increases, the amount of penalizing also increases; therefore the magnitude of the penalty term affects model complexity and number of eliminated features directly.

4.2.3. Decision Trees and Aggregation Algorithms

4.2.3.1. Decision Trees

Decision Trees isa machine learning method used in both regression and classification problems. It works by dividing heterogeneous data into homogeneous subgroups according to certain variables. This dividing process produces decision rules at the end. The main advantages of decision trees are that they're easy to interpret, they can work with both categorical and numerical variables. Although it requires low pre-processing, it's not able to work with missing values.

In order to decide where to divide or which variable should be use to create a new node or become a leaf, decision trees calculates impurity for all variables. There are two main functions to calculate: GINI impurity and entropy.

$$\boldsymbol{GINI\ Impurity = 1 - \sum_{i=1}^{n} (p_i)^2}$$

Formula 10

Where p_i is the probability of being class i.

$$\boldsymbol{Entropy = \sum_{i=1}^{n} -p_i \, log_2 \, p_i}$$

Formula 11

Similar to the GINI, entropy is another metric to measure heterogeneity. The main goal is to reach a 0 GINI value to stop dividing the tree.

4.2.3.2. Bagging

Bagging is a combination of the words bootstrap and aggregation. Just like the Boosting method, Bagging is an another method to aim prevent over-fitting and increase model's predictive performance. Bootstrap aggregation combine the models that builded by re-sampling. (Breiman, 1996) Because of this combining method, Bagging is a one of the Ensemble machine learning methods alongside Boosting. The main difference between them is that the Boosting method Works sequantially when the Bagging works parallel. To sum up, in Boosting created trees are dependent, but in Bagging they're independent from each other.

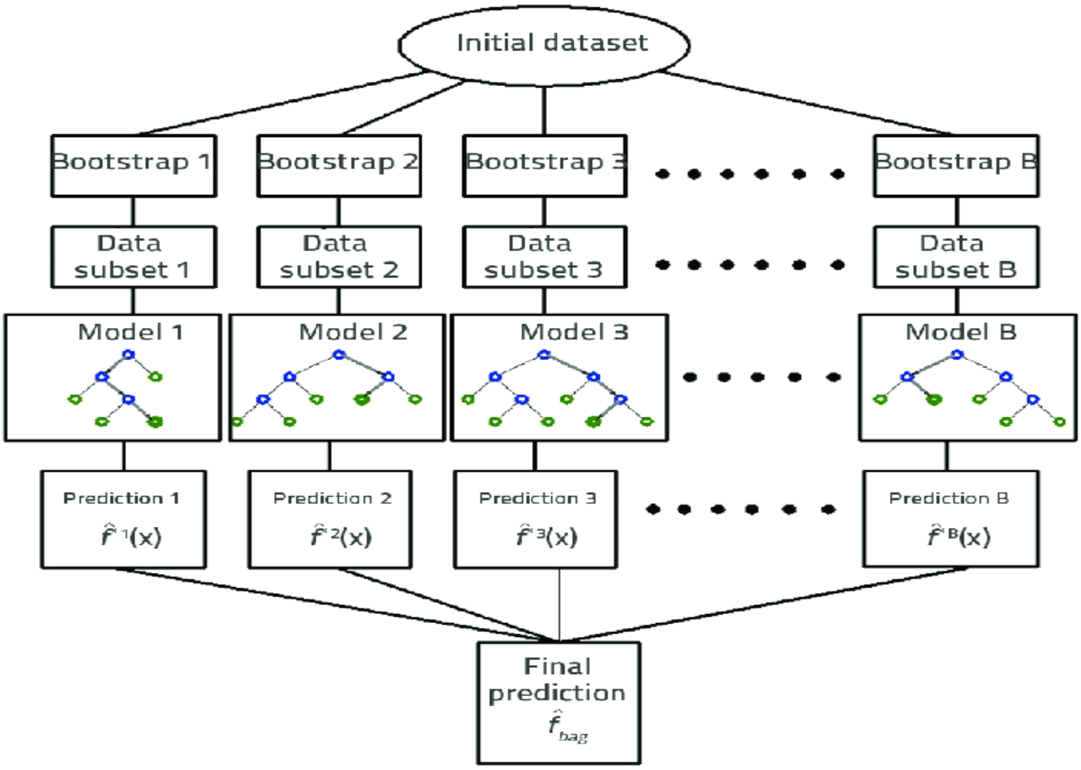


Figure 6: Bagging Schema

4.2.3.3. Random Forest

Random Forest is a type of decision trees, it can be also used in regression problems as well as classification problems. The main difference between Random Forest and decision trees (Classification and Regression Trees) is that random forest uses bootstrap aggregation. This approach makes random forest more unbiased but also more complex. This is because random forest creates randomness while training the

trees. Instead of looking for a single variable to produce a new leaf, it searchs for a variable from the random variable subset. Therefore, it does not only choose the data randomly, it also chooses variables randomly.

Because random forests builds multiple trees from different parts of the data and with different variables, it prevents over-fitting compared to a single decision tree.

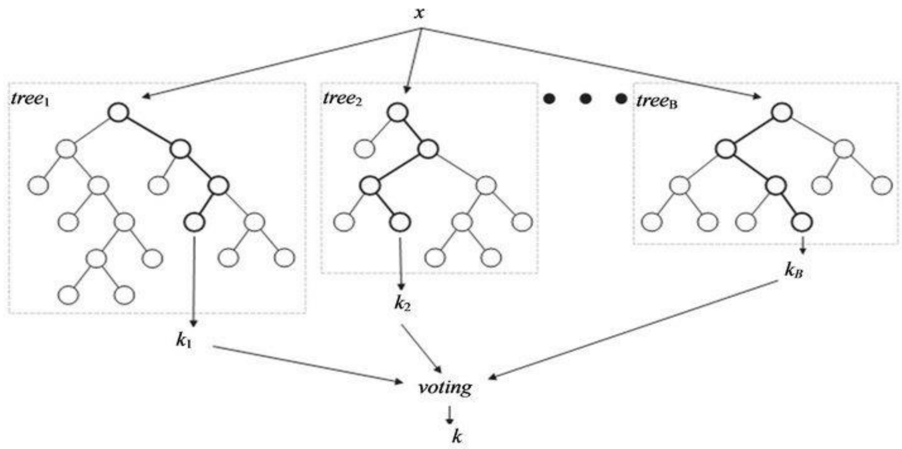


Figure 7: Random Forest Schema

4.2.3.4. Gradient Boosting

The Gradient Boosting is an "Additive Modeling" method that iteratively reduces the prediction error by establishing new models based the residuals of the previous models. It uses the Gradient Descent method, which minimizes the error by using a differentiable loss function.

There are several superior features of the Gradient Boosting method, it is able to operate with both categorical and continous data very well. And it's dealing with missing data itself without any manual preprocessing, which is a very time saving feature.

When the initial model is setting, the model makes is predictions by taking the average value of the dependent variable, based on the residuals. Then, the model produces other predictions with the lowest residuals. And after the new residuals are calculated, it repeats the same process. Thanks to this approach, one should expect that every next prediction will be better than the previous one. This iterative process keeps going until the loss function reaches it's minimum level (global minimum) or satisfy a determined threshold. See the minimization of cost function with gradient descent below.

Although it makes sense to take average of the dependent variables in regression, it's not same with the classification problems. In classification, the model needs a probability transformation to calculate residuals and create new trees by using them.

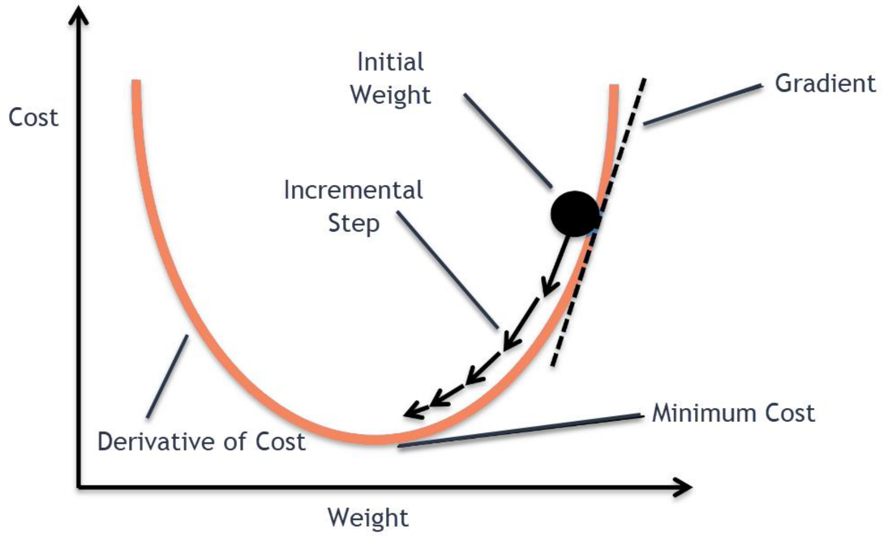


Figure 8: Gradient Descent Cost Minimization Method

The log(odds) formula for default is shown below:

$$\boldsymbol{log(odds) = \frac{Default}{Non - Default}}$$

Formula 12

If we convert this equation into an equation that takes log(odds) as input variable and gives probability outputs between 0 and 1, we obtain:

$$\boldsymbol{P(Default) = \frac{e^{log\,(odds)}}{1 + e^{log\,(odds)}}}$$

Formula 13

To obtain pseudo residuals/errors, subtraction of probabilty from the actual classes should occur, which is very simple,

$$\boldsymbol{Pseudo\ Residuals = Observed\ Values - Predicted\ Probability}$$

Formula 14

Note that, “Observed Values” only get values between 0 and 1. To produce value of the new trees , one more transformation is needed,, which is;

$$\boldsymbol{Output\ Value} = \frac{\sum \boldsymbol{Pseudo\ Residuals\ \ For\ A\ Single\ Leaf}}{\sum[\boldsymbol{Previous\ Probs.} * (\mathbf{1} - \boldsymbol{Previous.Probs.})]}$$

Formula 15

After the output value has been obtained, log(odds) predictions will be produced, to compute these values "Output Value" will be scaled by "learning rate". Which is a another topic that will be explained in the following sections of this study:

$$\boldsymbol{log(odds)\ Prediction} = \boldsymbol{Initial\ Tree} + (\boldsymbol{Learning\ Rate} * \boldsymbol{Output\ Value})$$

Formula 16

Finally, leafs are updated, but not in terms of probability, they're still log(odds). Lastly, these log(odds) should be converted into probabilities just like the initial case.

$$\boldsymbol{P(Default)} = \frac{\boldsymbol{e}^{\boldsymbol{log(odds)}}}{\mathbf{1} + \boldsymbol{e}^{\boldsymbol{log(odds)}}}$$

Formula 17

This was a very simplistic way to explain Gradient Boosting Classifiers, to have a more comprehensive and mathematical understanding about Gradient Boosting, please see "Greedy Function Approximation: A Gradient Boosting Machine" (2001) by Jerome H. Friedman.

4.2.3.5. Extreme Gradient Boosting (XGBoost)

XGBoost is another type of model that using Gradient Boosting method. Just like the other Boosting based methods it can perform with missing values without any preprocessing. It reaches the given max depth value of the trees and start pruning backwards when necessary.

XGBoost is considered to be the fastest and the most predictive of the gradient boosting-based models, which has been on the rise in the recent years. It minimizes the lost function with same approach with the Gradient Descent but XGBoost uses a regularization method. That's why it is so immune to over-fitting. (Chen, 2016)

$$\boldsymbol{L(\emptyset)} = \sum_{\boldsymbol{i}} \boldsymbol{l(\hat{y}_i, y_i)} \ + \sum_{\boldsymbol{k}} \Omega(\boldsymbol{f_k})$$

Formula 18

Is the cost function for XGBoost classifier, where Ω equals,

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda||w||2$$

Formula 19

Where w is the penalizer term. For more detailed explanation please see, "XGBoost: A Scalable Tree Boosting System" by Tianqi Chen.

4.2.3.6. Light GBM

Light GBM is another Gradient Boosting method which has also been quite popular in the recent years. It was developed in 2017 to increase the predictive power and training speed of XGBoost by Microsoft. It's compatible with parallel learning and GPU usage.

The most important feature that distinguishes it from other Gradient Boosting methods is the tree growing method that it adopts. In other Gradient Boosting methods trees build by using level-wise approach. Level-wise strategy aims to preserve the balance of the tree, to achieve the model that continues to search for the best split until the tree is become evenly splitted. This approach also called "horizontal growth".

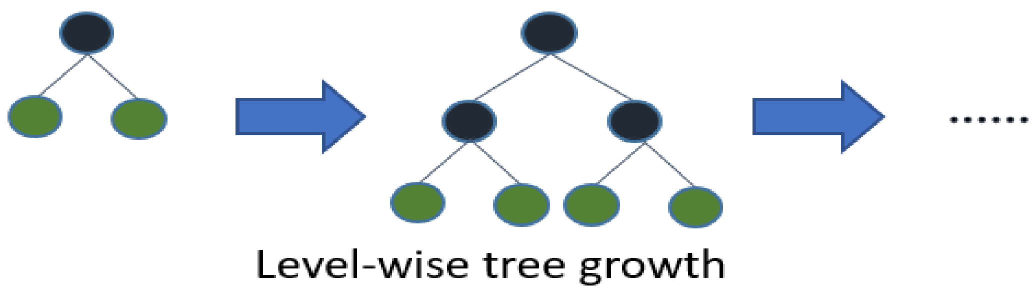


Figure 9: Horizontal Growth Approach Schema

Opposite to that, Light GBM adopts leaf-wise approach, which is found at the leafs that minimize loss function and keep splitting from them without considering the other leafs in same-stage. This asymmetrical growth also called "vertical growth".

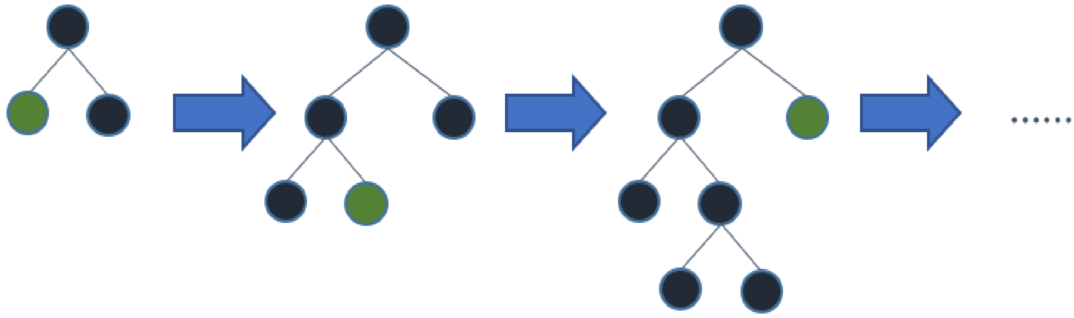


Figure 10: Vertical Growth Approach Schema

Because of leaf-wise growth, model training occurs faster, but trees are more sensitive to learning. This makes Light GBM more likely to suffers from over-fitting, specially in small datasets. To prevent that, learning hyperparameters of Light GBM should adjusted carefully.

4.2.3.7. CATBOOST (Categorical Boosting)

Cat Boost is an another Gradient Boosting based algorithm. It was created by Yandex in 2018 to provide performance development while working with categorical data. Like the other algorithms, Cat Boost can operate with both categorical and numerical data. In addition to that, Cat Boost provides auto encoding for categorical features. This makes it significantly easier to work with categorical features. Cat Boost especially be used in this study as it is claimed to give better results, especially when working with categorical variables.

4.2.4. Model Tuning/Hyperparameter Methods

Model tuning is an optimization process of Machine Learning models to find hyperparameters in order to obtain the maximum performance of the model. Unlike parameters, hyperparameters are not determined during mod training. Hyperparameter optimization is considered as an important factor in building an effective machine learning model.

There are too many different hyperparameters in machine learning (especially in tree-structured models) and deep learning models. These hyperparameters may differ in each model due to differences in model structures. This diversity creates challenges to adjust these hyperparameters manually, especially in more complex models. Therefore this automatization on model tuning is really crucial because it reduces the user efforts to the minimum level. We'll discuss different methods used for hyperparameter search in this section.

4.2.4.1. GridSearchCV

One of the most commonly used methods among hyper-parameter optimization methods is GridSearch CV. It is a quite slow algorithm that uses brute force method to search between combinations. It requires grid values to determine where to search for optimal values. GridSearch CV uses cross-validation technique to evaluate each trial/output that it produces.

Cross-validation is a model evaluation method that divides the dataset into k folds, and takes k-1 folds as train data and the other fold as test data. This dividing and model training process keeps going until the all combinations hav been used. Generally, the average of these trials is calculated to obtain a certain evaluation metric.

Figure 11: Fold Cross Validation Schema

*Source: (**Johar Aatqb,2019**)*

4.2.4.2. Randomized Search

Randomized search is a another hyper-paremeter optimization technique, which is similar with the GridSearch. They are the two most commonly used optimazation techniques. The key difference between them that makes randomized search more efficient is that randomized search randomly chooses the trials and decides the best parameters, when GridSearch tries all the grid. Because GridSearch is an exhaustive searching method, randomized search is more preferable, especially when dealing with big data. (Bergstra & Bengio, Random Search for Hyper-Parameter Optimization, 2012)

4.2.3.3. Bayesian Hyper-parameter Optimization

Bayesian hyper-parameter optimization is a more advanced technique that uses previous evaluations in order to improve performance and searching speed. There are many developed hyper-parameter optimization techniques based on this method. Bayesian optimization technique creates probability models of the loss functions and searches for the best hyper-parameters. After calculating of loss function with these parameters, the process is repeated until it reaches the pre-defined max iteration number.

Model	**Method**	**Hyper-Parameters**	**H**
LOGISTIC REGRESSION	**-**	**-**	**-**
LASSO LOGISTIC REGRESSION	**GridSearchCV**	**C (Penalizer)**	**0.01**
Light GBM	**Optuna**	**max_depth**	**6**
		n_estimators	**565**
		learning_rate	**1.0017229232072383e-08**
CART	**Optuna**	**max_depth**	**3**
		max_features	**4**
		min_samples_split	**11**

RANDOM FOREST	**Optuna**	**n_estimators**	**100**
		max_features	**6**
		min_samples_split	**2**
XGBOOST	**Optuna**	**max_depth**	**5**
		n_estimators	**348**
		learning_rate	**0.007011120300802943**
CATBOOST	**Optuna**	**max_depth**	**2**
		n_estimators	**589**
		learning_rate	**0.018858977807385702**

Table 2: Hyper-parameters of the Models

In this study, Optuna which is an algorithm that takes advantage of sequantial model based optimization (SMBO) is used. Which is another bayesian based optimization method. In order to have a more comprehensive understanding please see "Algorithms for Hyper-Parameter Optimization" by Bergstra & Bengio (2011).

5. MODEL VALIDATION

In this section, performance metrics and model evaluating process will be explained. In credit risk management, there is a branch called “Model Validation”, main duty of this branch is evaluate model’s performance and provide model compliance with regulatory obligations. Models that used in this study also be evaluated in this section.

5.1. Confusion Matrix

Confusion matrix is a square matrix that compares predicted and observed results. In classification problems, confusion matrix is a useful method to visualize classification results of the model. Consufion matrixes represent metrics such as true positive, false positive, true negative and false negative.

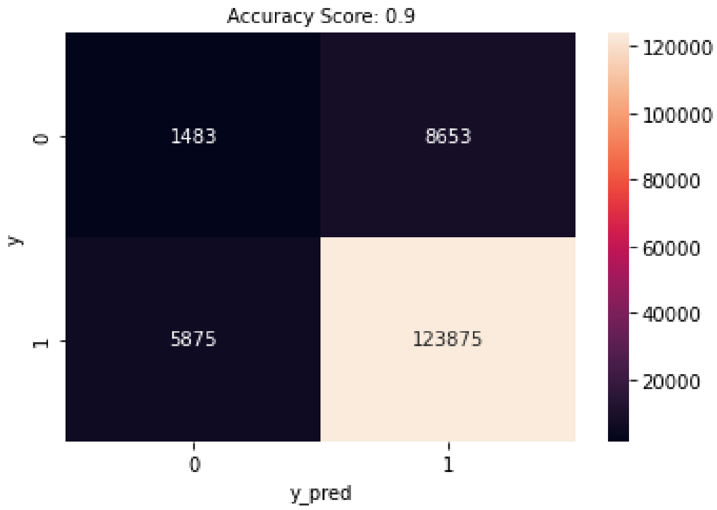


Figure 12: Confusion Matrix of XGBoost Model

Based on these values, common used metrics can be created. These metrics are shown below:

Accuracy

Accuracy represents the total succesful predictions of the model. It’s the most common metric to evaluate classification models. But it’s not a fair evaluation metric in unbalanced datasets.

$$\boldsymbol{Accuracy = \frac{True\ P + True\ N}{True\ P + True\ N + False\ P + False\ N}}$$

Formula 20

Precision

Precision measures the true positive accuracy. It is an important metric when cost of the false positives are high.

$$\boldsymbol{Precision = \frac{True\ P}{True\ P + False\ P}}$$

Formula 21

Recall

Recall also represents the true positive accuracy but in regards to a set of all positive predictions, "precision" solely highlights the corrects positive predictions. On the other hand, for examining the missed positive predictions, "recall" is the indicator in use.

$$\boldsymbol{Recall = \frac{True\ P}{True\ P + False\ N}}$$

Formula 22

F1 Score

Another important metric, the F1 score represents the harmonic mean of the precision and recall values. The fact that it contains all error metrics is effective in playing an important role in model selection.

$$\boldsymbol{F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}}$$

Formula 23

Model	Accuracy	Precision	Recall	F1 Score	Number of True Negatives (Defaults)
Logistic Regression(WoE)	**.7578**	**.9487**	**.7788**	**. 8554**	**4648**
Lasso Logistic Regression(WoE)	**.7445**	**.9497**	**. 7651**	**. 8474**	**4811**
Decision Tree Classifier	**. 9007**	**.9296**	**.9661**	**.9475**	**647**
Random Forest Classifier	**.8359**	**.9401**	**.8791**	**.9086**	**2873**
XGBoost Classifier	**.8961**	**.9347**	**.9547**	**.9446**	**1483**
LightGBM Classifier	**.8438**	**.9330**	**.8959**	**.9141**	**1792**
CatBoost Classifier	**.9041**	**.9330**	**.9659**	**.9492**	**1147**
XGBoost Classifier(WoE)	**.7095**	**.9517**	**.7236**	**.8221**	**5347**
CatBoost Classifier(WoE)	**.8207**	**.9417**	**.8599**	**.8990**	**3213**

Table 3: Accuracy, Precision, Recall, F1 Scores of Various Models

5.2. Brier Score

Brier score is a simple metric that is used for measuring the loss of predicted probabilities.

$$\boldsymbol{Brier\ Score} = \frac{\mathbf{1}}{\boldsymbol{n}} \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{n}} (\boldsymbol{f_i} - \boldsymbol{d_i})^{\mathbf{2}}$$

Formula 24

In the equation above, f_i stands for the predicted probability of the certain observation and d_i is the real value of the observation.

5.3. Area Under The Receiver Operating Characteristic Curve (AUC)

The Receiver Operating Characteristic curve is graphed with the False Positive ratio on the x-axis and the True Positive ratio on the y-axis. The ROC curve is formed by plotting each point where the True Positive and False Positive ratios intersect. While visualizing the ROC curve, a linear line is drawn at an angle of 45 degrees to represent a model that has no informational value or a random classifier. We can say that the ROC Curve of a model with high classification power will move away from this linear line or the area under it will increase.

AUC refers to the area under the ROC curve and refers to how well the classifier distinguishes between default and non-default. When the classification power increases, area under the curve also increases.

$$\boldsymbol{AUC} = \int_{\mathbf{0}}^{\mathbf{1}} \boldsymbol{TPR}\,(\boldsymbol{FPR})\,\boldsymbol{dFPR}$$

Formula 25

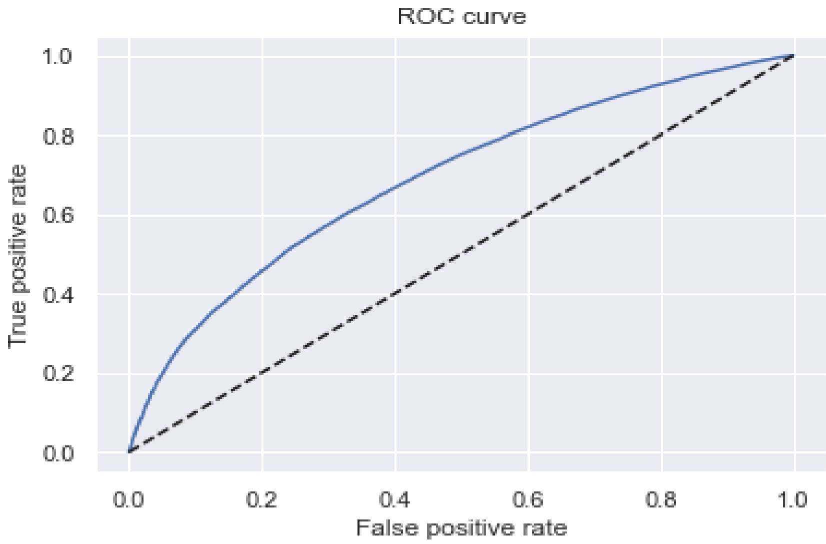


Figure 13: ROC Curve of the LASSO Logistic Model

5.4. GINI Curve

GINI curve can be plotted in order to evaluate model performance, but it carries the same information with AUC score. Therefore, GINI can be calculated using AUC value. Which is simply,

$$\boldsymbol{GINI = (AUC * 2) - 1}$$

Formula 26

5.5. Kolmogorov-Smirnov (KS)

Kolmogorov-Smirnov statistic is used to calculate the maximum difference between cumulative true positive and cumulative false positive rates. When the value distinguishing ability of the model increases, KS statistic also increases. KS statistic is a commonly used metric in credit risk model validation. Please see "The Kolmogorov-Smirnov Test for Goodness of Fit" (1951) by Frank Jay Massey.

Model	Brier Score	AUC	KS
Logistic Regression(WoE)	**.0714**	**.6906**	**.2780**

Lasso Logistic Regression(WoE)	**.0718**	**.6891**	**.2759**
Decision Tree Classifier	**.0695**	**.5821**	**.1204**
Random Forest Classifier	**.0679**	**.6731**	**.2540**
XGBoost Classifier	**.0683**	**.7022**	**.2975**
Light GBM Classifier	**.0821**	**.6212**	**.1297**
CatBoost Classifier	**.0653**	**.6963**	**.2879**
XGBoost Classifier(WoE)	**.0743**	**.6870**	**.2719**
CatBoost Classifier(WoE)	**.0691**	**.6850**	**.2721**

Table 4: Brier AUC and KS Values

6. FINDINGS

In this study, we built various types of models to compare their predictive powers. After probability of default values were created by these models, our threshold to be non-default was set as .80 or higher. Then different performance metrics were calculated for each model.

According to the results in “Table 5: Accuracy, Precision, Recall, F1 Scores of Various Models”, in terms of accuracy, two models with top performances were CatBoost and Decision Tree Classifier. While CatBoost classifier had 90.41% accuracy which is more succesful than the any other models in our study, Decision Tree Classifer showed 90.07% accuracy. These models also had greatest recall values which were 96.61% for Decision Tree Classifier and 96.59% for CatBoost Classifier. But these models were both insufficient to estimate the true defaulters.

In terms of precision, high scorer 4 models are the models that use WoE grouped data. Which are XGBoost Classifier (WoE), Logistic Regression, LASSO Logistic Regression and CatBoost Classifier (WoE). We can say that this models have highest ratio in true positive against false positives. Therefore these models are the best models that estimated true defaulters.

According to the results in “Table 6: Brier AUC and KS Values” best AUC values belonged to XGBoost Classifier, CatBoost Classifier and Logistic Regression. XGBoost Classifier had the highest AUC value with 70.22%, CatBoost Classifier had the second highest with 69.63% and Logistic Regression with 69.06%. These models also had higher Kolmogorov-Smirnov values as expected, telling us that these models were the best models for distinguishing between the default and non-default classes.
When we look at the brier loss score, models with lowest values were CatBoost Classifier, Random Forest Classifier and XGBoost Classifier. This tells us said models produce closest probabilities to actual classes.

Even though the the results in Table 3 and Table 4 are quiet detailed and diverse, some of them might be misleading. For instance, accuracy metric is the most direct and easily interpreted performance metric, but in an unbalanced dataset like ours, accuracy might not reflect the model performance. As we discussed in the data section, our data has 10.9% default rate, so if we would have estimated all the observations as a non-defaults, then we would have obtained 89.9% accuracy already. Therefore, true positives and false positives can be more appropriate performance metrics to make a comparison. To measure this, when the AUC and Kolmogorov-Smirnov values were observed, we can say that XGBoost Classifier outperformed all the models in our study. And also Catboost Classifier, which is known by it’s performance on working with categorical data, proved that it has the second highest distiguishing ability between defaults and non-defaults with 69.63% AUC and 28.79% KS.

7. CONCLUSION

Logistic regression models are used with Weight of Evidence groupings to create a sustainable and stable model for credit risk modeling. Because WoE grouping make data processing more costly, it is important to consider whether there is a better method or not. In this study, linear classifiers and alternative classifiers were compared.

In this study, variable selection was made by WoE groupings and IV. Threshold for being a good borrower was determined as a .80 probability and higher. After 22 variables were chosen, LASSO shrinked some of them. Logistic regression used all of them even with the variables with low coefficients. These 22 variables were also used by machine learning models but without WoE groupings, except for the variables that include missing values. Tree based models also eliminated or remained useless for some of the variables. Feature importances graphs will be shown in the “Appendix” section.

According to the results, in terms of accuracy, machine learning models outperformed the conventional models. But because of the unbalanced dataset, it’s not surprising to achieve a high accuracy rate. Because banks need to estimate defaulters, and reject their loan applications but also they have to keep lending to proper borrowers, they have to build a model that has a high distinguishing power between these two classes.

In terms of predicting more number of defaults correctly, linear models did predict certainly better compared to other models which are built on non WoE grouped data. Weight of evidence grouping penalized the bad borrowers very succesfuly for sure, which is very cruical since the main goal of credit risk management is to prevent loss. In order to compare WoE grouping data performance between conventional and tree based ML models, XGBoost and CatBoost used grouped data and XGBoost outperformed conventional models in terms of number of true default predictions, while CatBoost couldn’t. We can certainly say that XGBoost was the most succesful model in this study. But also we can say that CatBoost has distiguished the defaulters and non defaulters better than the linear models because of higher Area Under The

Curve and Kolmogorov-Smirnov values, therefore providing better true positive and false positive rates.

According to the number of true negatives (defaulters/bad borrowers), we can say that best models were the models that used data with Weight of Evidence groupings. Thus, we also interpreted that WoE groupings provide better classification results for defaulters. We can say that this manual process is quite helpful to prevent credit losses.

To sum up, gradient boosting classifiers such as XGBoost and CatBoost can be used instead of linear models with or without WoE if the legal regulations in the banking sector changes. Of course, operational costs of the models should be considered in this process. Also, in order to achieve a more comprehensive and detailed comparison, more complex models like Neural Networks or Voting Classifiers can be used. Additionally, one should consider macroeconomic features in order to increase predictive power. Lastly, increasing the number of WoE grouped variables for linear models and providing more variables to tree based machine learning models would be helpful.

REFERENCES

Alshari, H. & Saleh, Y & Odabas, A. (2021). *Comparison of Gradient Boosting Decision Tree Algorithms for CPU Performance*. Erciyes Tip Dergisi. 157-168.

Bergstra, J., Bardenet, Ré., Bengio, Y. & Kégl, B. (2011). *Algorithms for Hyper-parameter Optimization*. Proceedings of the 24th International Conference on Neural Information Processing Systems (p./pp. 2546--2554), USA: Curran Associates Inc.. ISBN: 978-1-61839-599-3

Breiman, L. (1996). *Bagging Predictors*. Machine Learning, 24, 123-140.

Dionne, G. *Structured Finance, Risk Management, and the Recent Financial Crisis* (October 13, 2009).

Dionne, G. (2013*). Risk Management: History, Definition, and Critique*. ERN: Other Microeconomics: Decision-Making under Risk & Uncertainty.

Expected Credit Loss. (2022, February 18). Open Risk Manual, . Retrieved 23:41, June 3, 2022 from https://www.openriskmanual.org/wiki/index.php?title=Expected_Credit_Loss&oldid=27765.

Friedman, J.H. (2001) *Greedy function approximation: A gradient boosting machine*.. Ann. Statist. 29 (5) 1189 - 1232, October

Halim, S. & Humira,Y. (2014). *Credit Scoring Modelling*, Jurnal Teknik Industri, Vol. 16, No. 1, 17-24.

Kovacevic, M & Ivanisevic, N & Petronijevic, P & Despotovic, V. (2021). *Construction cost estimation of reinforced and prestressed concrete bridges using machine learning*. 73. 1-13. 10.14256/JCE.2738.2019.

Kuhn, M., & Johnson, K. (2018). *Applied predictive modeling*. Springer. 495-500.

Massey, F. J. (1951). *The Kolmogorov-Smirnov Tests for Goodness of Fit*. Journal of American Statistical Association, 46.

Nikolay G. (2022, March 9). *Credit Risk Modeling in Python 2022*. (365 Careers) Retrieved from Udemy: https://www.udemy.com/course/credit-risk-modeling-in-python/

Oktay Fırat, Ü. (1996). *REGRESYON ANALİZİNDE BAĞIMSIZ DEĞİŞKENLERİN BELİRLENMESİNİ SAĞLAYAN ALGORİTMALAR* . Öneri Dergisi , 1 (5) , 49-54 . DOI: 10.14783/maruoneri.703374

Polena, M. (2017). *Performance Analysis of Credit Scoring Models on Lending Club Data*. Master's Thesis, Charles University Faculty of Social Sciences, Prague.

Shekhar, S & Bansode, A & Salim, A (2022). *A Comparative study of Hyper-Parameter Optimization Tools.*

Tang et al. Financial Innovation (2020) 6:36 https://doi.org/10.1186/s40854-020-00197-y (al., 2020)

Tianqi, C. Carlos, G. (2016*). XGBoost: A Scalable Tree Boosting System.* San Francisco, CA, USA.

Wang H, Xu Q, Zhou L (2015) *Large Unbalanced Credit Scoring Using Lasso-Logistic Regression Ensemble*. PLOS ONE 10(2): e0117844.

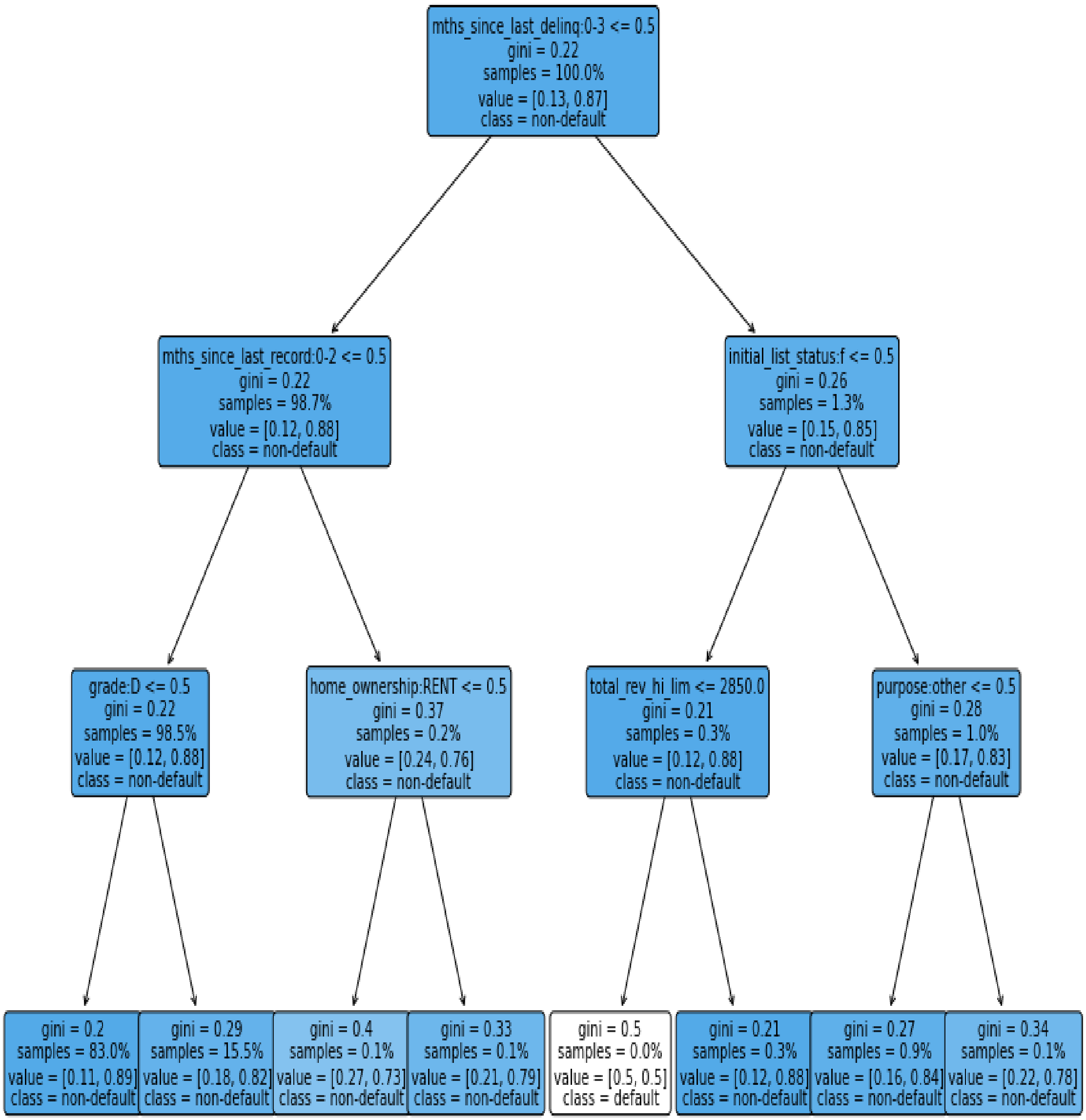
Yang, S & Berdine, G. (2017). *The receiver operating characteristic (ROC) curve*. The Southwest Respiratory and Critical Care Chronicles. 5. 34. 10.12746/swrccc.v5i19.391.

Yang, Li & Shami, A. (2020). *On Hyperparameter Optimization of Machine Learning Algorithms: Theory and Practice*, 16-27.

Zhang, C. & Cao, L & Romagnoli, A. (2018). *On the feature engineering of building energy data mining. Sustainable Cities and Society.* 39. 10.1016/j.scs.2018.02.016.

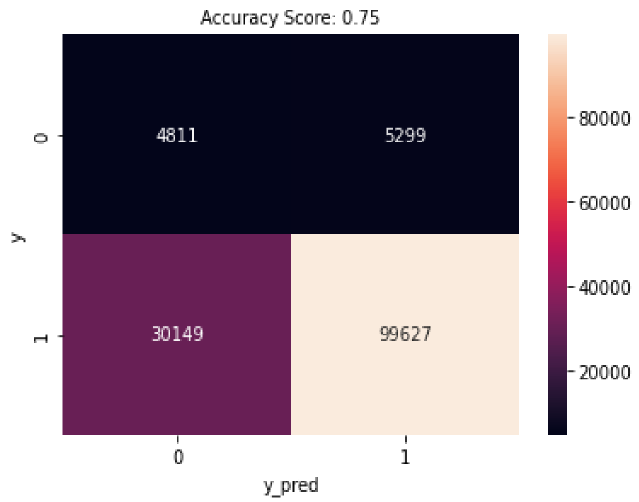
APPENDIX

APPENDIX 1. Decision Tree Classifier (Representative)

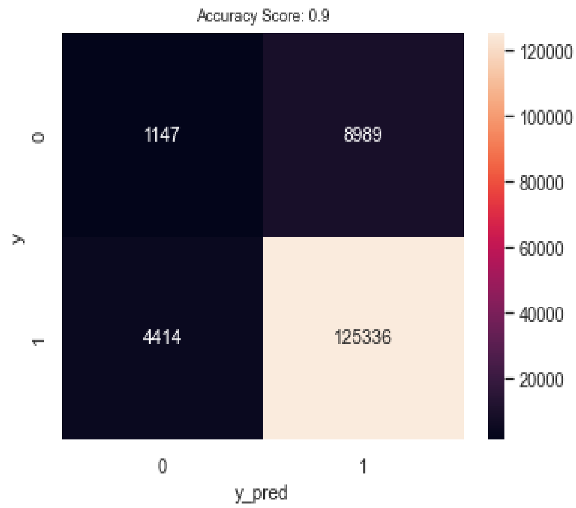


APPENDIX 2. Confusion Matrixes

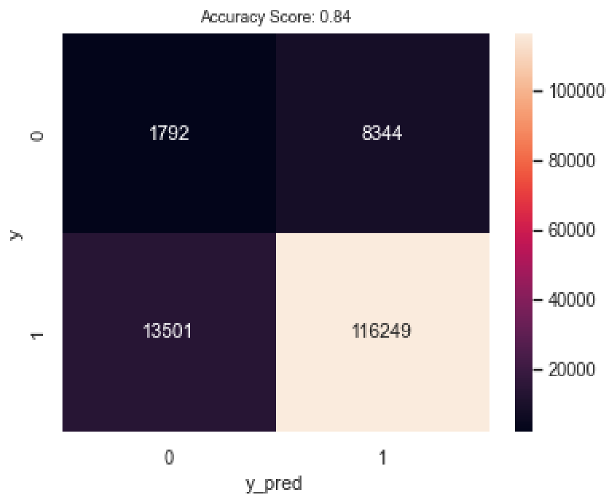
LASSO



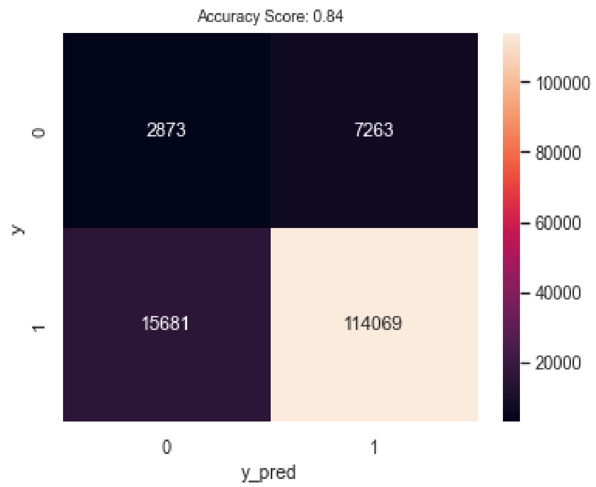
CatBoost



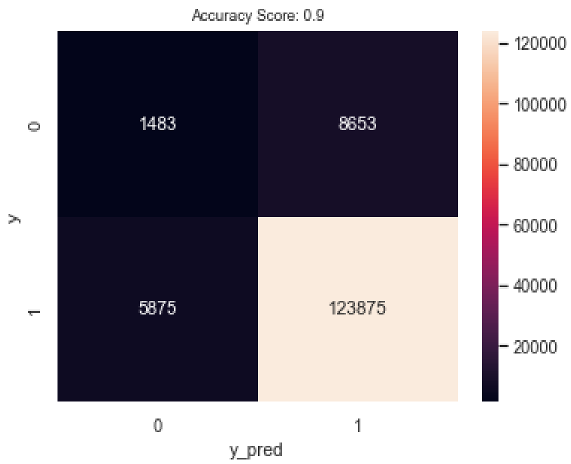
Light GBM



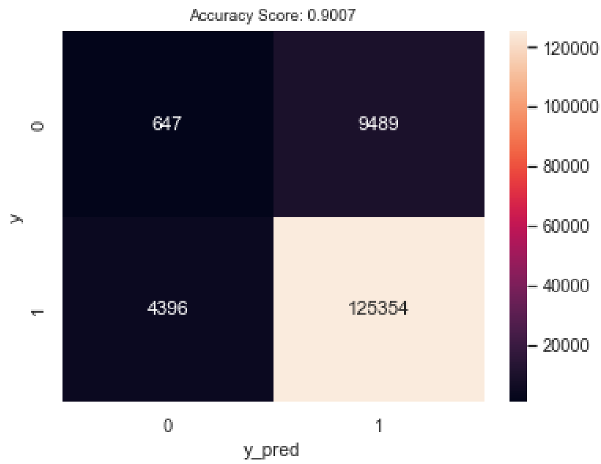
Random Forest



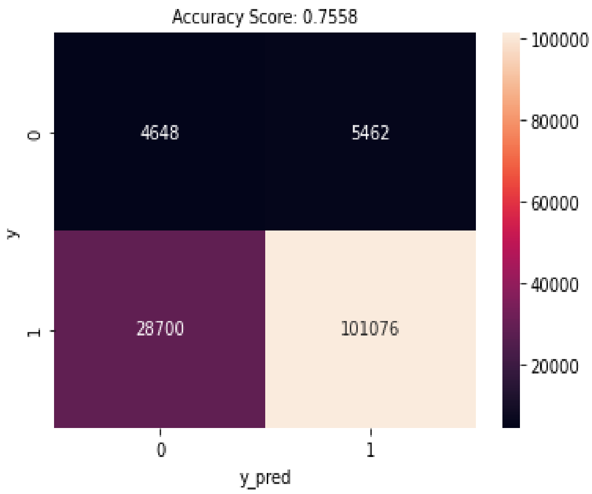
XGBoost Classifier



Decision Tree Classifier

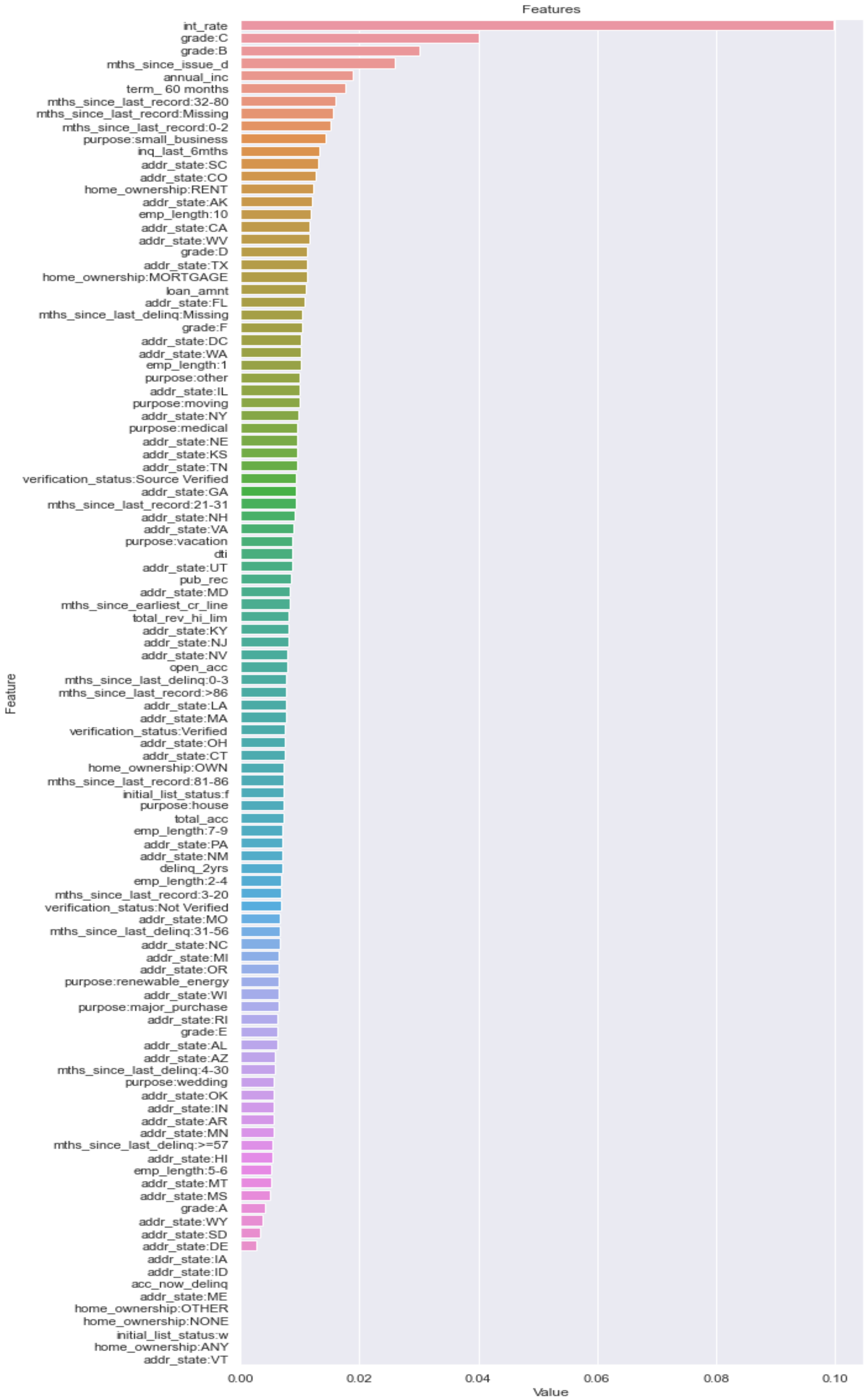


Logistic Regression

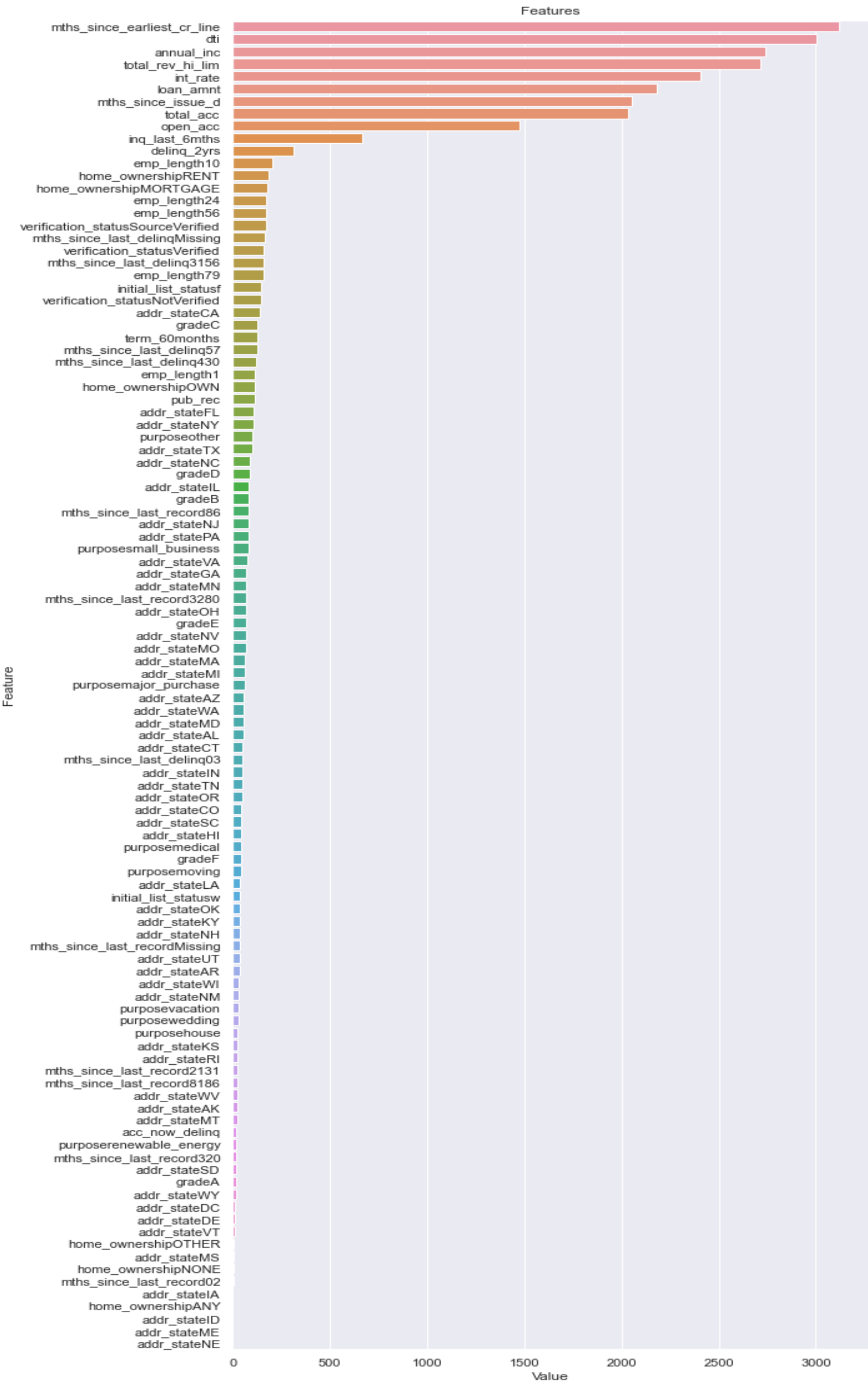


APPENDIX 3. Feature Importances

XGBoost Classifier Feature Importance



Light GBM Classifier Feature Importance



APPENDIX 4. Variables

grade: External evaluated loan grade

home_ownership: Home ownership.

addr_state: The state provided by the borrower in the loan application

verification_status: Is the income of the applicant has been proved?

purpose: Purpose

initial_list_status: The initial listing status of the loan.

term: Term

mths_since_issue: How many months has been passed since last credit issue

int_rate: Interest rate

mths_since_earliest_cr_line: The month the borrower's earliest reported credit line was opened

delinq_2yrs: Number of delinquencies in last two years

inq_last_6mths: Number of credit inquiries in last six months

open_acc: The number of open credit lines of the customer.

pub_rec: Public record

total_acc: The total number of credit lines currently in the borrower's credit file.

acc_now_delinq: The number of accounts on which the borrower is now delinquent.

total_rev_hi_lim: Total revolving high credit limit

annual_inc: Annual income

dti: Debt to income ratio(monthly)

mths_since_last_delinq: The number of months since the borrower's last delinquency.

mths_since_last_record: The number of months since the last public record.

-loan_amnt: Loan amount.